Discrete-1.2.70

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Problem

Solve the differential equation:

$$\frac{dy}{dx} = x^2 + 2y$$

with initial condition $y = 1$ at $x = 0$.

Options:

(a) $y = \frac{2}{3}x^2 + 4x$

(b) $y = \sqrt{\frac{2}{3}x^3 + 4x + k}$

(c) $y = \frac{2}{3}x^3 - 4x + k$

(d) $y = \frac{2}{3}x^3 + 4x + k$

Step 1: Rewrite in Standard Form

Rearrange:

$$\frac{dy}{dx} - 2y = x^2$$

This is a **first-order linear ODE** of the form:

$$\frac{dy}{dx} + P(x)\,y = Q(x)$$

where

$$P(x) = -2, \qquad Q(x) = x^2$$

Step 2: Integrating Factor

The integrating factor is:

$$\mu(x) = e^{\int P(x)\,dx} = e^{\int -2\,dx} = e^{-2x}$$

Multiply both sides by $\mu(x) = e^{-2x}$:

$$e^{-2x}\frac{dy}{dx} - 2e^{-2x}y = x^2e^{-2x}$$

The left-hand side becomes a perfect derivative:

$$\frac{d}{dx}\left(y\,e^{-2x}\right) = x^2e^{-2x}$$

Step 3: Integrate Both Sides

$$\int d\left(y\, e^{-2x}\right) = \int x^2\, e^{-2x}\, dx$$

Using integration by parts twice on the right:

$$\int x^2 e^{-2x}\, dx = -\frac{x^2}{2} e^{-2x} - \frac{x}{2} e^{-2x} - \frac{1}{4} e^{-2x} + C$$

Therefore:

$$y\, e^{-2x} = -\frac{x^2}{2} e^{-2x} - \frac{x}{2} e^{-2x} - \frac{1}{4} e^{-2x} + C$$

Dividing by e^{-2x}:

$$y = Ce^{2x} - \frac{x^2}{2} - \frac{x}{2} - \frac{1}{4}$$

Step 4: Apply Initial Condition

Apply $y = 1$ at $x = 0$:

$$1 = C\,e^0 - 0 - 0 - \frac{1}{4}$$

$$1 = C - \frac{1}{4} \implies C = \frac{5}{4}$$

Final Solution

$$y = \frac{5}{4}e^{2x} - \frac{x^2}{2} - \frac{x}{2} - \frac{1}{4}$$

Verification

Check ODE: $y = \frac{5}{4}e^{2x} - \frac{x^2}{2} - \frac{x}{2} - \frac{1}{4}$

$$\frac{dy}{dx} = \frac{5}{2}e^{2x} - x - \frac{1}{2}$$

$$x^2 + 2y = x^2 + \frac{5}{2}e^{2x} - x^2 - x - \frac{1}{2} = \frac{5}{2}e^{2x} - x - \frac{1}{2} \checkmark$$

Check IC: $y(0) = \frac{5}{4} - 0 - 0 - \frac{1}{4} = 1 \checkmark$

Conclusion

$$y = \frac{5}{4}e^{2x} - \frac{x^2}{2} - \frac{x}{2} - \frac{1}{4}$$

RK4 Method: Overview

The ODE is $f(x, y) = x^2 + 2y$.

The **Runge-Kutta 4th order** update rule:

$$k_1 = h\, f(x_n,\ y_n)$$

$$k_2 = h\, f\left(x_n + \tfrac{h}{2},\ y_n + \tfrac{k_1}{2}\right)$$

$$k_3 = h\, f\left(x_n + \tfrac{h}{2},\ y_n + \tfrac{k_2}{2}\right)$$

$$k_4 = h\, f(x_n + h,\ y_n + k_3)$$

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Starting from $x_0 = 0,\ y_0 = 1$, step size $h = 0.25$.

RK4 Numerical Results

x	RK4	Exact	Error
0.0000	1.000000	1.000000	0.00e+00
0.2500	1.654378	1.654652	2.73e-04
0.5000	2.771898	2.772852	9.54e-04
0.7500	4.693406	4.695861	2.46e-03
1.0000	7.980764	7.986320	5.56e-03
1.2500	13.560169	13.571867	1.17e-02
1.5000	22.958397	22.981921	2.35e-02
1.7500	38.692244	38.738065	4.58e-02
2.0000	64.910489	64.997688	8.72e-02

Observation

Error grows with x due to the exponential nature of the solution, but remains small relative to the solution magnitude.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

def f(x, y):
    return x**2 + 2*y

def rk4(f, x0, y0, xf, h):
    xs, ys = [x0], [y0]
    x, y = x0, y0
    while x < xf - 1e-10:
        k1 = f(x, y)
        k2 = f(x + h/2, y + h/2 * k1)
        k3 = f(x + h/2, y + h/2 * k2)
        k4 = f(x + h, y + h * k3)
```

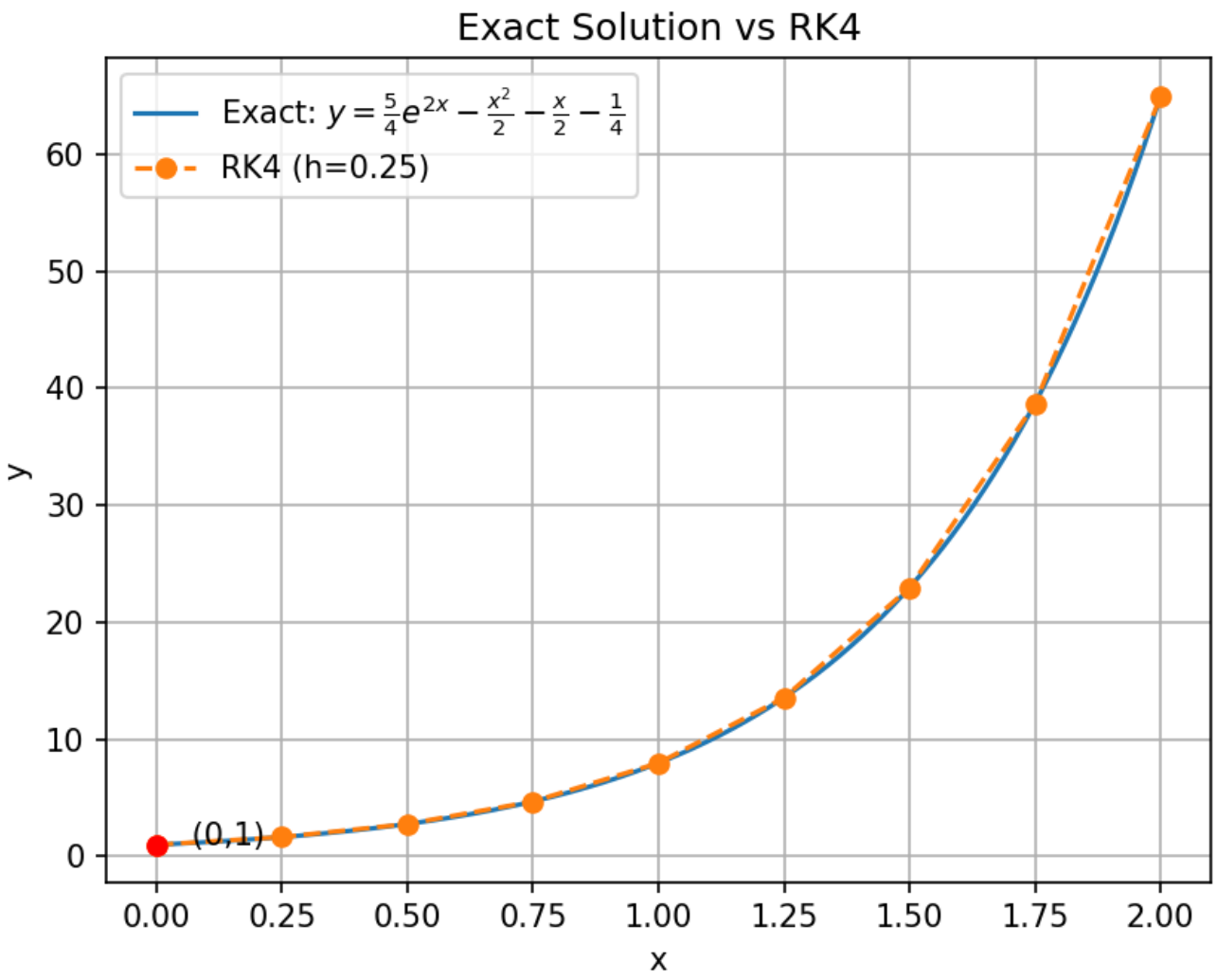
Python Code

```
 y = y + h/6*(k1 + 2*k2 + 2*k3 + k4)
        x = x + h
        xs.append(x); ys.append(y)
    return np.array(xs), np.array(ys)
x0, y0, xf, h = 0.0, 1.0, 2.0, 0.25
xs, ys_rk4 = rk4(f, x0, y0, xf, h)

x_plot = np.linspace(0, 2, 500)
C = 5/4
y_plot = C*np.exp(2*x_plot) - x_plot**2/2 -
   x_plot/2 - 1/4

plt.figure()
plt.plot(x_plot, y_plot,
    label=r'Exact: $y=\frac{5}{4}e^{2x}-\frac{x
       ^2}{2}-\frac{x}{2}-\frac{1}{4}$')
```

Python Code

```
plt.plot(xs, ys_rk4, 'o--', label='RK4 (h=0.25)')
plt.scatter([0], [1], color='red', zorder=5)
plt.text(0.05, 1, ' (0,1)')
plt.title('Exact Solution vs RK4')
plt.xlabel('x'); plt.ylabel('y')
plt.legend(); plt.grid()
plt.savefig('plot2.png', dpi=150)
```

Graph of the Solution



- Exact solution curve and RK4 points on same plot